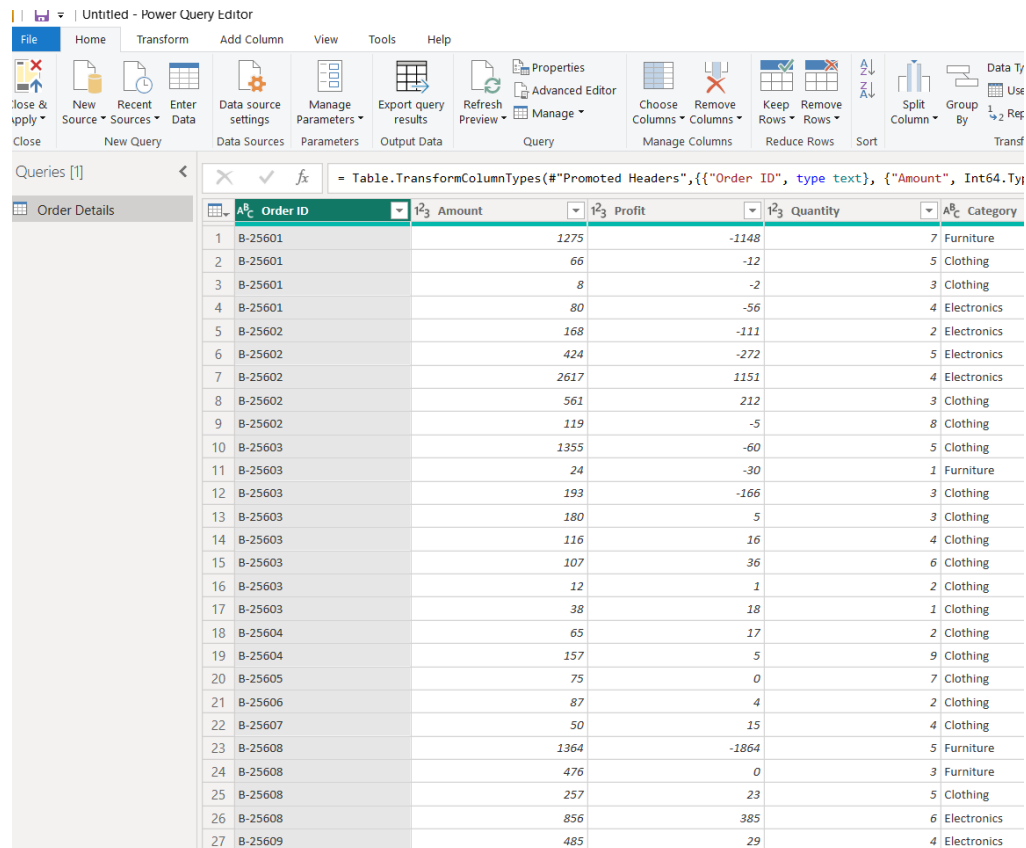


Power BI Assignment 1 – Data Transformation & Data Modeling

As a Data Analyst, you will work with an E-commerce sales dataset to perform data preparation and

modeling using Power BI. The dataset contains multiple related files representing order-level, product-level, and sales target information.

The focus of this assignment is on data import, transformation, cleansing, aggregation, and relationship modelling using Power BI and Power Query Editor, while ensuring consistency, accuracy, and business relevance of the prepared data model.



The screenshot displays the Power Query Editor interface. The top ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Transform tab is active, showing various data manipulation options like 'Choose Columns', 'Remove Columns', 'Keep Rows', 'Remove Rows', 'Sort', 'Split Column', 'Group By', and 'Data Type'. The main area shows a table with the following data:

	Order ID	Amount	Profit	Quantity	Category
1	B-25601	1275	-1148	7	Furniture
2	B-25601	66	-12	5	Clothing
3	B-25601	8	-2	3	Clothing
4	B-25601	80	-56	4	Electronics
5	B-25602	168	-111	2	Electronics
6	B-25602	424	-272	5	Electronics
7	B-25602	2617	1151	4	Electronics
8	B-25602	561	212	3	Clothing
9	B-25602	119	-5	8	Clothing
10	B-25603	1355	-60	5	Clothing
11	B-25603	24	-30	1	Furniture
12	B-25603	193	-166	3	Clothing
13	B-25603	180	5	3	Clothing
14	B-25603	116	16	4	Clothing
15	B-25603	107	36	6	Clothing
16	B-25603	12	1	2	Clothing
17	B-25603	38	18	1	Clothing
18	B-25604	65	17	2	Clothing
19	B-25604	157	5	9	Clothing
20	B-25605	75	0	7	Clothing
21	B-25606	87	4	2	Clothing
22	B-25607	50	15	4	Clothing
23	B-25608	1364	-1864	5	Furniture
24	B-25608	476	0	3	Furniture
25	B-25608	257	23	5	Clothing
26	B-25608	856	385	6	Electronics
27	B-25609	485	29	4	Electronics

Row Limiting & Data Types:

- Restrict the List of Orders table to the first 500 rows.
- Set the Order Date column to the Date data type.
- Change the Amount and Target columns to fixed decimal numbers.

Power Query Editor interface showing a table with columns: Month of Order Date, Category, and Target. The formula bar shows a transformation: `= Table.TransformColumnTypes(#"Promoted Headers",{{"Month of Order Date", type date}, {"Category", type text}, {"Target", Int64.Type}})`.

	1	2	3
	Month of Order Date	Category	Target
1	4/18/2026	Furniture	10400
2	5/18/2026	Furniture	10500
3	6/18/2026	Furniture	10600
4	7/18/2026	Furniture	10800
5	8/18/2026	Furniture	10900
6	9/18/2026	Furniture	11000
7	10/18/2026	Furniture	11100
8	11/18/2026	Furniture	11300
9	12/18/2026	Furniture	11400
10	1/19/2026	Furniture	11500
11	2/19/2026	Furniture	11600
12	3/19/2026	Furniture	11800
13	4/18/2026	Clothing	12000
14	5/18/2026	Clothing	12000
15	6/18/2026	Clothing	12000
16	7/18/2026	Clothing	14000
17	8/18/2026	Clothing	14000
18	9/18/2026	Clothing	14000
19	10/18/2026	Clothing	16000
20	11/18/2026	Clothing	16000
21	12/18/2026	Clothing	16000
22	1/19/2026	Clothing	16000
23	2/19/2026	Clothing	16000
24	3/19/2026	Clothing	16000
25	4/18/2026	Electronics	9000
26	5/18/2026	Electronics	9000
27	6/18/2026	Electronics	9000
28	7/18/2026	Electronics	9000

8	-2	3	Clothing	Hankerchief
80	-56	4	Electronics	Electronic Games
12	1	2	Clothing	Hankerchief
38	18	1	Clothing	Kurti
65	17	2	Clothing	T-shirt

×

Keep Top Rows

Specify how many rows to keep.

Number of rows

500

OK

Cancel

\$ Target
10,400.00
10,500.00
10,600.00
10,800.00
10,900.00
11,000.00
11,100.00
11,300.00
11,400.00
11,500.00
11,600.00
11,800.00
12,000.00
12,000.00
12,000.00
14,000.00
14,000.00
14,000.00
16,000.00
16,000.00
16,000.00
16,000.00
16,000.00
16,000.00
9,000.00
9,000.00
9,000.00
9,000.00

Format the CustomerName column to proper case to ensure consistent capitalization.

Column Creation:

○ Merge the City and State columns to create a new column named Location in the format:

City, State

○ Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)

1/4/2018	Pearl	Maharashtra	Pune
3/4/2018	Jahan	Madhya Pradesh	Bhopal
3/4/2018			
5/4/2018			
6/4/2018			
6/4/2018			
8/4/2018			
9/4/2018			
9/4/2018			
11/4/2018			
12/4/2018			
12/4/2018			
13-04-2018			
15-04-2018			
15-04-2018			
17-04-2018	Sagar	Nagaland	Kohima
18-04-2018	Manju	Andhra Pradesh	Hyderabad
18-04-2018	Ramesh	Gujarat	Ahmedabad
20-04-2018	Sarita	Maharashtra	Pune

Merge Columns

Choose how to merge the selected columns.

Separator
--None--

New column name (optional)
location

OK Cancel

Table.AddColumn("#Capitalized Each Word", "location", each Text.Combine([State], [City]), type text)						
Order ID	Order Date	CustomerName	State	City	location	
1	B-25601	1/4/2018	Bharat	Gujarat	Ahmedabad	GujaratAhmedabad
2	B-25602	1/4/2018	Pearl	Maharashtra	Pune	MaharashtraPune
3	B-25603	3/4/2018	Jahan	Madhya Pradesh	Bhopal	Madhya PradeshBhopal
4	B-25604	3/4/2018	Divsha	Rajasthan	Jaipur	RajasthanJaipur
5	B-25605	5/4/2018	Kasheen	West Bengal	Kolkata	West BengalKolkata
6	B-25606	6/4/2018	Hazel	Karnataka	Bangalore	KarnatakaBangalore
7	B-25607	6/4/2018	Sonakshi	Jammu and Kashmir	Kashmir	Jammu and KashmirKashmir
8	B-25608	8/4/2018	Aarushi	Tamil Nadu	Chennai	Tamil NaduChennai
9	B-25609	9/4/2018	Jitesh	Uttar Pradesh	Lucknow	Uttar PradeshLucknow
10	B-25610	9/4/2018	Yogesh	Bihar	Patna	BiharPatna
11	B-25611	11/4/2018	Anita	Kerala	Thiruvananthapuram	Kerala Thiruvananthapuram
12	B-25612	12/4/2018	Shrichand	Punjab	Chandigarh	PunjabChandigarh
13	B-25613	12/4/2018	Mukesh	Haryana	Chandigarh	HaryanaChandigarh
14	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla	Himachal PradeshSimla
15	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok	SikkimGangtok
16	B-25616	15-04-2018	Kanak	Goa	Goa	GoaGoa
17	B-25617	17-04-2018	Sagar	Nagaland	Kohima	NagalandKohima
18	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad	Andhra PradeshHyderabad
19	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad	GujaratAhmedabad
20	B-25620	20-04-2018	Sarita	Maharashtra	Pune	MaharashtraPune
21	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal	Madhya PradeshBhopal
22	B-25622	22-04-2018	Monisha	Rajasthan	Jaipur	RajasthanJaipur

Conditional Column:

○ Add a conditional column named Profit Status based on:

■ Profit < 0 → Loss

■ Profit = 0 → Break-Even

■ Profit > 0 → Profit

Custom Column

Add a column that is computed from the other columns.

New column name

Custom Margin

Custom column formula

= [Profit] / [Amount]

Learn about Power Query formulas

Available columns

Order ID

Amount

Profit

Quantity

Category

Sub-Category

<< Insert

✓ No syntax errors have been detected.

OK

Cancel

Amount

Profit

Quantity

Category

Sub-Category

Custom Margin

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
profit status

	Column Name	Operator	Value		Output
If	Profit	is less than	0	Then	loss
Else If	Profit	equals	0	Then	Breakeven

Add Clause

Else
profit

OKCancel

Table.AddColumn(#"Changed Type1", "profit status", each if [Profit] < 0 then "loss" else if [Profit] = 0 then "Breakeven" else "profit")						
	Profit	Quantity	Category	Sub-Category	Custom Margin	profit status
1	1275	-1148	Furniture	Bookcases	-90.04%	loss
2	66	-12	Clothing	Stole	-18.18%	loss
3	8	-2	Clothing	Hankierchief	-25.00%	loss
4	80	-56	Electronics	Electronic Games	-70.00%	loss
5	168	-111	Electronics	Phones	-66.07%	loss
6	424	-272	Electronics	Phones	-64.15%	loss
7	2617	1151	Electronics	Phones	43.98%	profit
8	561	212	Clothing	Saree	37.79%	profit
9	119	-5	Clothing	Saree	-4.20%	loss
10	1355	-60	Clothing	Trousers	-4.43%	loss
11	24	-30	Furniture	Chairs	-125.00%	loss
12	193	-166	Clothing	Saree	-86.01%	loss
13	180	5	Clothing	Trousers	2.78%	profit
14	116	16	Clothing	Stole	13.79%	profit
15	107	36	Clothing	Stole	33.64%	profit
16	12	1	Clothing	Hankierchief	8.33%	profit
17	38	18	Clothing	Kurti	47.37%	profit
18	65	17	Clothing	T-shirt	26.15%	profit
19	157	5	Clothing	Saree	3.18%	profit
20	75	0	Clothing	Saree	0.00%	Breakeven
21	87	4	Clothing	Shirt	4.60%	profit
22	50	15	Clothing	Leggings	30.00%	profit
23	1364	-1864	Furniture	Tables	-136.66%	loss
24	476	0	Furniture	Chairs	0.00%	Breakeven
25	257	23	Clothing	Hankierchief	8.95%	profit
26	856	385	Electronics	Printers	44.98%	profit
27	485	29	Electronics	Electronic Games	5.98%	profit

Merging Data (Joins)

- Merge the List of Orders and Order Details tables using Order ID.
- Name the merged output table as Orders Data.

Advanced Editor

Use First Row as Headers

Append

Merge

Select a table and matching columns to create a merged table.

Order ID

Order ID	Order Date	CustomerName	State	City	location
B-25601	1/4/2018	Bharat	Gujarat	Ahmedabad	GujaratAhmedabad
B-25602	1/4/2018	Pearl	Maharashtra	Pune	MaharashtraPune
B-25603	3/4/2018	Jahan	Madhya Pradesh	Bhopal	Madhya PradeshBhopal
B-25604	3/4/2018	Divsha	Rajasthan	Jaipur	RajasthanJaipur
B-25605	5/4/2018	Kasheen	West Bengal	Kolkata	West BengalKolkata

Order Details

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Custom Margin	profit status
B-25601	1275	-1148	7	Furniture	Bookcases	-90.04%	loss
B-25601	66	-12	5	Clothing	Stole	-18.18%	loss
B-25601	8	-2	3	Clothing	Hankerchief	-25.00%	loss
B-25601	80	-56	4	Electronics	Electronic Games	-70.00%	loss
B-25602	168	-111	2	Electronics	Phones	-66.07%	loss

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

⚠ Select the same number of columns from both visible tables to continue.

OK Cancel

Sorting and Filtering Data

- Sort records in the Orders Data table by Order Date in descending order.
- Apply filters to analyze orders from a specific state (e.g., Tamil Nadu).

Monospaced ☐ Column distribution ☐ Always allow ☐ Show whitespace ☒ Column profile ☒ Column quality ☒ Go to Column ☐ Advanced Editor ☐ Query Dependencies

Data Preview Columns Parameters Advanced Dependencies

Table.AddColumn(#"Changed Type1", "profit status", each if [Profit] < 0 then "loss" else if [Profit] = 0 then "Breakeven" else "profit")

Order ID	Amount	Profit	Quantity	Category	Sub-Category	% Custom Mar
1 B-25601	1275	-1148	7	Furniture	Bookcases	
2 B-25601	66	-12	5	Clothing	Stole	
3 B-25601	8	-2	3	Clothing	Hankerchief	
4 B-25601	80	-56	4	Electronics	Electronic Games	
5 B-25602	168	-111	2	Electronics	Phones	
6 B-25602	424	-272	5	Electronics	Phones	
7 B-25602	2617	1151	4	Electronics	Phones	
8 B-25602	561	212	3	Clothing	Saree	
9 B-25602	119	-5	8	Clothing	Saree	
10 B-25603	1355	-60	5	Clothing	Trousers	

Column statistics

Count	500
Error	0
Empty	0
Distinct	167
Unique	81
Empty string	0
Min	B-25601
Max	B-25767

Value distribution

B-25656	
B-25757	
B-25751	
B-25651	
B-25650	
B-25753	
B-25750	
B-25755	
B-25653	
B-25655	
B-25754	

Query Settings

PROPERTIES

Name

Order Details

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Kept First Rows

Added Custom

Changed Type1

Added Conditional Column

Table.Sort(#"Promoted Headers1",{"8/4/2018", Order.Descending}))

B-25608	8/4/2018	Aarushi	Tamil Nadu	Chennai	Tamil NaduChennai
1 B-26008	9/2/2019	Kalyani	Tamil Nadu	Chennai	Tamil NaduChennai
2 B-25698	23-06-2018	Amisha	Tamil Nadu	Chennai	Tamil NaduChennai
3 B-26081	22-03-2019	Aarushi	Tamil Nadu	Chennai	Tamil NaduChennai
4 B-25788	21-09-2018	Dinesh	Tamil Nadu	Chennai	Tamil NaduChennai
5 B-25860	15-11-2018	Akshay	Tamil Nadu	Chennai	Tamil NaduChennai
6 B-26018	14-02-2019	Aarushi	Tamil Nadu	Chennai	Tamil NaduChennai
7 B-25716	11/7/2018	Surabhi	Tamil Nadu	Chennai	Tamil NaduChennai

Grouping and Aggregating Data

● Duplicate the Order Details table and perform the following:

○ Count of Order ID

○ Average Profit by Category

○ Total Amount by Sub-Category

● Duplicate the Sales Target table and aggregate the total Target by Month of order date.

Table	Any Column	Text Column
eries [8]	✕ ✓ <i>f_x</i> = Table.Group("#Inserted Date", {"Target"}, {"total target", each List.Sum(	
Order Details	📅 \$ Target 1.2 total target	
Sales target	1 10,400.00 10400	
List of Orders	2 10,500.00 10500	
Order ID	3 10,600.00 10600	
Order Details (2)	4 10,800.00 10800	
Order Details (3)	5 10,900.00 10900	
Order Details (4)	6 11,000.00 11000	
Sales target (2)	7 11,100.00 11100	
	8 11,300.00 11300	
	9 11,400.00 11400	
	10 11,500.00 11500	
	11 11,600.00 11600	
	12 11,800.00 11800	
	13 12,000.00 36000	
	14 14,000.00 42000	
	15 16,000.00 144000	
	16 9,000.00 81000	

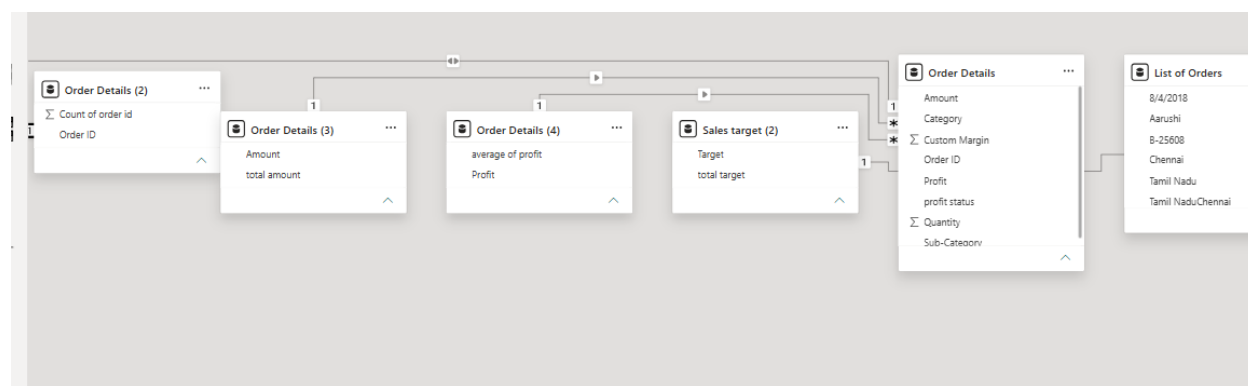
Grouping and Aggregating Data

- Duplicate the Order Details table and perform the following:
 - Count of Order ID
 - Average Profit by Category
 - Total Amount by Sub-Category
- Duplicate the Sales Target table and aggregate the total Target by Month of order date.

A ^B _C Order ID	1 ² ₃ Count of order id
B-25601	1
B-25602	1
B-25603	1
B-25604	1
B-25605	1
B-25606	1
B-25607	1
B-25608	1
B-25609	1
B-25610	1
B-25611	1
B-25612	1
B-25613	1
B-25614	1
B-25615	1
B-25616	1
B-25617	1
B-25618	1
B-25619	1
B-25620	1
B-25621	1
B-25622	1
B-25623	1
B-25624	1
B-25625	1
B-25626	1
B-25627	1
B-25628	1

1 ² ₃ Amount	1.2 total amount
1275	1275
168	336
1355	1355
65	65
75	150
87	261
50	50
1364	1364
485	485
1076	1076
160	160
259	259
1603	1603
494	494
68	204
42	126
305	305
362	362
353	353
193	386
233	233
534	1068
53	53
26	52
97	97
46	92
55	55
45	45

Month of Order Date	Category	\$ Target	Month Name
4/18/2026	Furniture	10,400.00	April
5/18/2026	Furniture	10,500.00	May
6/18/2026	Furniture	10,600.00	June
7/18/2026	Furniture	10,800.00	July
8/18/2026	Furniture	10,900.00	August
9/18/2026	Furniture	11,000.00	September
10/18/2026	Furniture	11,100.00	October
11/18/2026	Furniture	11,300.00	November
12/18/2026	Furniture	11,400.00	December
1/19/2026	Furniture	11,500.00	January
2/19/2026	Furniture	11,600.00	February
3/19/2026	Furniture	11,800.00	March
4/18/2026	Clothing	12,000.00	April
5/18/2026	Clothing	12,000.00	May
6/18/2026	Clothing	12,000.00	June
7/18/2026	Clothing	14,000.00	July
8/18/2026	Clothing	14,000.00	August
9/18/2026	Clothing	14,000.00	September
10/18/2026	Clothing	16,000.00	October
11/18/2026	Clothing	16,000.00	November
12/18/2026	Clothing	16,000.00	December
1/19/2026	Clothing	16,000.00	January
2/19/2026	Clothing	16,000.00	February
3/19/2026	Clothing	16,000.00	March
4/18/2026	Electronics	9,000.00	April
5/18/2026	Electronics	9,000.00	May
6/18/2026	Electronics	9,000.00	June



Data Modeling:

